

CROSSSAFE · DESIGN PRECIS

CrossSafe design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ 17ef4fd		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices adopted for TRL 3 work (CRS-DDR-001), PIR gating, pilot light, 114.3 mm post, numbers from CRS-CAL-001, model and drawing CRS-DWG-001
0.2	2026-09-25	Amish Chadha	TRL 2 concept (how it works, components, first-order numbers, safety, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CrossSafe design precis

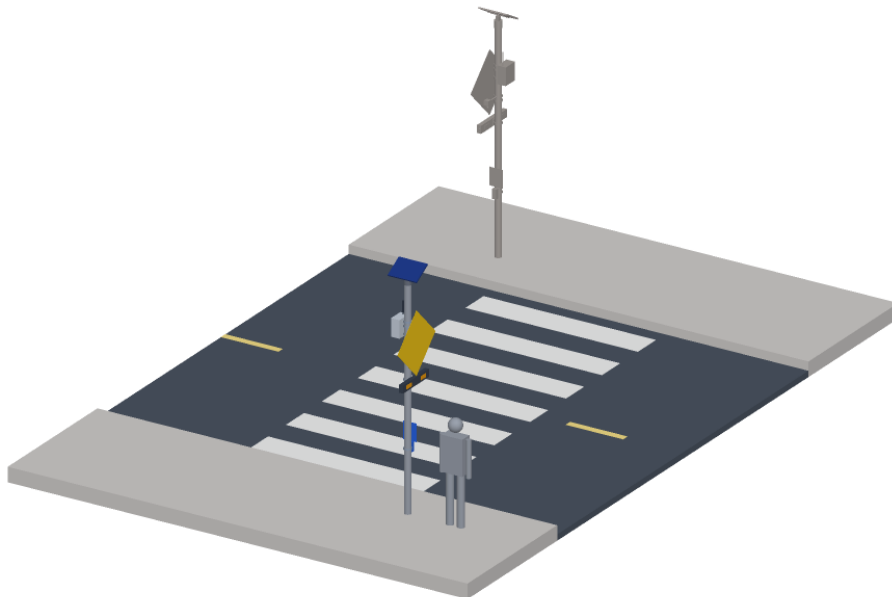
■ Summary

CrossSafe is a pair of pole-mounted, solar-powered warning beacons, one on each side of an unsignalized crossing. When a pedestrian presses the button or a small radar sensor sees someone waiting at the kerb, both assemblies flash two amber lights per face under the crossing warning sign for about 13 s, and a small pilot light confirms to the pedestrian that the warning is on. Each assembly runs from a 20 W panel and a 12.8 V 12 Ah LiFePO₄ battery, a low-power PIR sensor switches the radar on only when something moves nearby, and the two sides stay in step over a short LoRa link, so no trench or cable across the road is needed. The TRL 3 calculations (CRS-CAL-001) give 16.0 Wh/day of use at 300 activations, 7.7 days without sun, and \$318.00 in parts for one assembly on an existing pole. Heat in the enclosure and slip of the sign clamps are the open risks.

The concept follows the rectangular rapid flashing beacon (RRFB), which FHWA lists as a proven safety countermeasure ([FHWA](#)), and uses the US Interim Approval IA-21 as its reference for light size, flash pattern, night dimming and the optional pilot light ([FHWA IA-21](#)). It is an open reference design, not a certified traffic control device.

CrossSafe

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

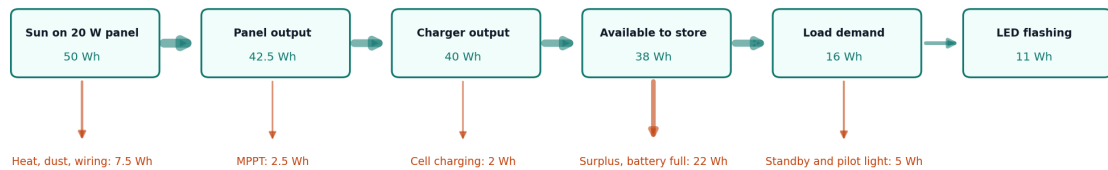
■ How it works

1. **Idle.** The controller sleeps with its radio listening, the button is armed, and a PIR sensor watches for movement near the kerb. When the PIR sees motion it switches on the radar, which watches a waiting zone of about 1.5 x 2 m; the radar goes off again when nothing has moved for a set time. Passive detection can be switched off where it is not allowed. The battery charges from the panel.
2. **Trigger.** A button press, or a person detected in the waiting zone for 1 s or more, wakes the controller.
3. **Synchronize.** The controller sends a short LoRa packet to the far-side assembly (and listens for the same from it). Both start flashing within 48 ms, or 244 ms if two packets are lost (CRS-CAL-001).
4. **Warn.** Two amber heads per face flash in the IA-21 wig-wag sequence, 75 sequences per minute, for the site's set time (crossing length / 0.9 m/s + 5 s, 12.8 s for 7 m), dimmed at night. A new trigger extends the time. The pilot light on the kerb end of the light bar confirms to the pedestrian that the warning is on, and the button gives a click and a tactile pulse.
5. **Log.** The controller counts activations and records faults and battery state. It stores no images or audio; the radar reports presence only.

Figure 2 shows the daily energy flow and Figure 3 the parts. The general arrangement is drawing CRS-DWG-001 (cad/drawings/), generated from the parametric model `cad/src/model.py` .

CrossSafe: daily energy per assembly, Wh per day (CRS-CAL-001 estimates: 2.5 peak sun hours, 300 activations of 20 s)

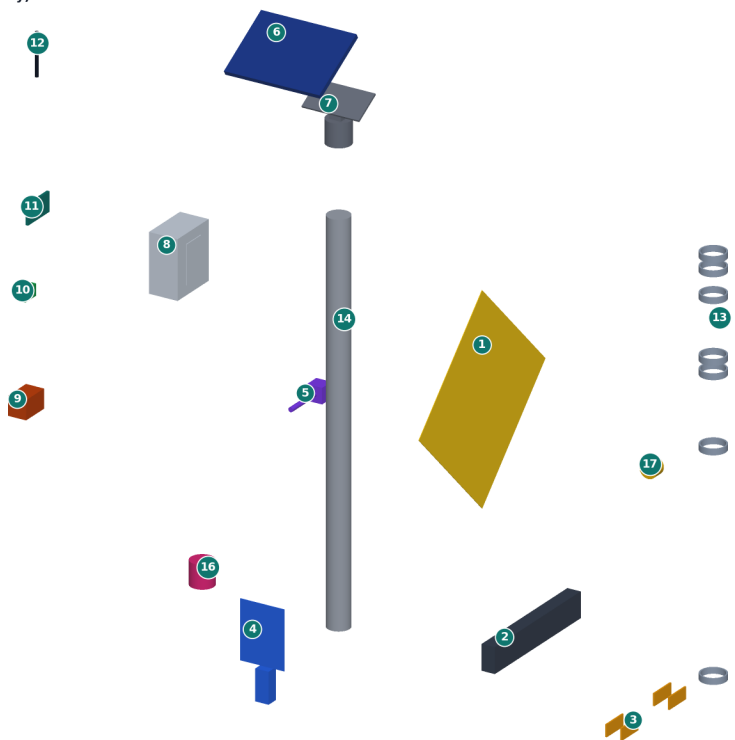
CONCEPT, NOT FOR FABRICATION



CrossSafe: exploded view (one assembly)

CONCEPT, NOT FOR FABRICATION

- 1 Crossing warning sign, 750 mm
- 2 Light bar housing, double sided
- 3 Amber LED heads (4)
- 4 Push-button station
- 5 Presence radar on arm
- 6 Solar panel, 20 W
- 7 Panel bracket
- 8 Enclosure, IP65
- 9 LiFePO4 battery 12.8 V 12 Ah
- 10 MPPT charge controller
- 11 Controller and radio board
- 12 Antenna
- 13 Pole clamps
- 14 Post, 114.3 mm (shortened in this view)
- 16 PIR wake sensor
- 17 Pedestrian pilot light



Post shortened, push button moved up, PIR (16) and pilot light (17) drawn at three times size for this view; numbers match bom/bom.csv

■ Main components

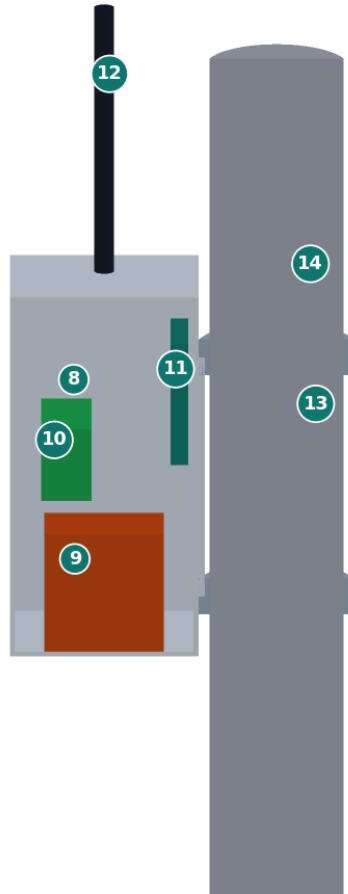
Table 1. Components of one assembly. Numbers match the exploded view and bom/bom.csv (line 15, wiring and protection, has no callout).

No.	Component	Concept choice
1	Crossing warning sign	750 mm diamond, retroreflective, local sign design
2	Light bar housing	Folded aluminium, double sided, 720 x 130 x 70 mm, bottom at 2.1 m
3	LED heads (4)	Amber, 140 x 62 mm lens (127 x 51 mm minimum), two per face, with daylight and night levels
4	Push-button station	Vandal-resistant piezo button with acknowledgement tone and tactile arrow, instruction plate, at about 1.0 m
5	Presence radar	24 GHz presence sensor on a 200 mm arm at 2.6 m, aimed at the waiting zone, powered only when the PIR sees motion
6	Solar panel	20 W monocrystalline, about 500 x 360 mm, tilted about 30 degrees toward the equator
7	Panel bracket	Steel top bracket on the post cap
8	Enclosure	IP65 polycarbonate, 150 x 260 x 300 mm, membrane vent, cable glands, 2.85 to 3.15 m, behind the post
9	Battery	LiFePO4 12.8 V 12 Ah (about 154 Wh) with internal BMS
10	Charge controller	Small MPPT controller with a LiFePO4 profile and low-temperature charge cut-off
11	Controller and radio board	FieldNode radio core (STM32WL-class LoRa module) in point-to-point mode, four LED drivers with night dimming, real-time clock, fault indicator
12	Antenna	Short whip on the enclosure roof
13	Pole clamps	Stainless band clamps with brackets
14	Post	Existing pole first; where a new post is needed, 114.3 x 3.6 mm galvanized steel, 3.7 m above the sidewalk, 1.8 m in a 500 mm footing
16	PIR wake sensor	Low-power passive infrared sensor under the radar; switches the radar on
17	Pedestrian pilot light	Small amber light on the kerb end of the light bar, where local rules allow

CrossSafe: cutaway of the pole-top enclosure

CONCEPT, NOT FOR FABRICATION

- 8 Enclosure, IP65
- 9 LiFePO4 battery 12.8 V 12 Ah
- 10 MPPT charge controller
- 11 Controller and radio board
- 12 Antenna
- 14 Post (section of)
- 13 Pole clamps



Enclosure cut on the pole axis; battery low, charger and controller above, antenna on the roof

■ Key design choices

Each choice below is adopted for TRL 3 work under Amish's 2026-09-25 instruction, open for his review (CRS-DDR-001).

- **RRFB-style amber beacons, not a signal (D4).** Amber warning lights with the standard sign ask drivers to yield; they do not claim to stop traffic, which keeps the device in a simpler approval class in most places. IA-21 is the baseline; the flash pattern is configurable for other jurisdictions.
- **Button plus passive detection, gated by a PIR (D2, D5).** The button is the reliable, accessible trigger; radar catches people who do not press it and can be switched off where passive detection is not allowed. The PIR keeps the radar off when nothing moves, which is what brings autonomy above 5 days. The radar reports presence only, which keeps images and audio off the device.
- **Radio sync instead of a cable (D6).** A LoRa point-to-point link avoids cutting the road. If the link drops, each side still flashes on its own trigger and reports a fault.
- **Own 12 V power system, FieldNode radio core (D7).** The LED load (up to about 26 W at the instant all four heads are lit) is far beyond the FieldNode power stage (6 W panel, 3.2 V cell), so CrossSafe uses a 12.8 V LiFePO4 battery with a built-in BMS. The controller reuses FieldNode's STM32WL-class radio and logging design. CellGuard, the lab's open BMS for 4 to 16 LiFePO4 cells, is a later option in place of the drop-in battery's closed BMS.

- **Existing poles first; 114.3 mm where a new post is needed (D3).** Band clamps fit 60 to 114.3 mm poles; the new-post variant meets the wind target with margin.
- **Pedestrian pilot light (D8).** IA-21 permits a small pilot light on the beacon or push button; it is included where local rules allow.
- **Electronics high on the pole.** The enclosure bottom at 2.85 m and the panel at the top deter theft and keep cables out of reach.

■ Key numbers (CRS-CAL-001)

The TRL 2 first-order estimates have been replaced by the TRL 3 calculation note CRS-CAL-001, which states every assumption and prints every figure from `docs/04-calcs/sizing.py`. The main results:

- **Flash time.** 12.8 s for a 7 m crossing; the energy case uses 20 s, enough for crossings up to 13.5 m.
- **Energy.** Each head is lit 25 % of each IA-21 sequence, so the LEDs use 10.99 Wh/day at 300 activations. Standby is 187 mW (4.49 Wh/day) with the PIR gating the radar. The daily load is 16.0 Wh, against 38.0 Wh stored in the worst month; break-even is 1.05 peak sun hours, and the battery refills from 20 % in 2.05 days at 5 peak sun hours.
- **Autonomy.** 122.9 Wh usable gives 7.7 days without sun, 5.4 days at -20 °C and 6.2 days at end of life.
- **Heat.** The enclosure runs 9.6 K (clean) to 14.7 K (dusty) above ambient in full sun, so charging stops above about 30 to 35 °C ambient and a dusty box reaches 64.7 °C at 50 °C ambient (R8 at risk). A ventilated shield would cut the rise to 3.9 K (proposed, awaiting Amish).
- **Latency and link.** 48 ms side to side, 244 ms with two retries; about 60 dB of link margin across the road with a bus in the way.
- **Wind.** A 40 m/s gust gives 3.35 kN·m at the base. The 114.3 x 3.6 mm post reaches 100 MPa, 42 % of S235 yield; an 88.9 x 4 mm post would reach 62 %. The sign clamps can slip in torsion at the assumed band tension (R9 at risk). A 500 mm footing needs about 1.8 m of embedment in clay (screening).
- **Cost.** \$318.00 per assembly on an existing pole (\$350 budget) and \$636.00 per crossing; \$388.00 and \$776.00 with new posts (see `bom/bom.csv`).

■ Safety

Safety: CrossSafe is a research prototype. A traffic control device must meet local standards and have road authority approval before use on a public road. A beacon that fails dark, flashes when no one is there, or gives pedestrians false confidence can cause harm; pedestrians must still check that drivers have stopped.

Safety: Lithium cells can overheat, vent and burn. Use LiFePO4 with a BMS, fuse the battery output close to the terminal, charge only between 0 and 45 °C cell temperature, and shade the enclosure. Never charge a damaged or swollen battery.

Safety: Installing on a pole next to live traffic is work at height and work in the road. Use trained crews, traffic management and fall protection as local rules require, keep clear of overhead power lines, and mount only with the asset owner's permission. A post, sign or panel that falls can kill. CRS-CAL-001 is a screening check only: clamp torsion is at risk, and a qualified engineer must check the post, clamps, footing or host pole to the local code before installation.

Safety: Sign and panel edges are sharp and heavy; handle with gloves and a second person.

■ Open questions

- [] Pilot jurisdiction and its rules for beacon color, flash pattern, sign design and mounting height.
- [] Measured LED head power and intensity for the chosen modules, and whether 6 W per head is enough in full sun.
- [] Radar false-trigger rate with passing pedestrians, cyclists, animals and rain.
- [] Radio range and reliability across the road with parked vehicles and buses in the path, and licence-free band rules in the pilot country.
- [] Enclosure temperature in full sun: a sun shield, or moving the enclosure under the panel (CRS-CAL-001, section C).
- [] Anti-rotation detail for the sign clamps and measured band tension (R9); host pole checks by the owner; local foundation design.
- [] Whether the pedestrian-facing confirmation light is allowed locally.